

Product Summary

You have a full-stack e-commerce platform: a branded storefront for customers plus an admin operations panel for store owners. It is built with Spring Boot, JWT security, JPA, Stripe integration, email templates, local/cloud image storage, and a React/MUI frontend.

What you are selling is essentially a ready-to-customize online store system with product/catalog management, checkout, order fulfillment, customer accounts, admin reporting, inventory controls, coupons, refunds, notifications, and demo data for showing the whole product.

Customer Storefront

Customers can:

- Browse products on the home/catalog page.
- Search products by text.
- Filter by category, price range, in-stock status, and rating.
- Sort by name, price low/high, price high/low, and newest.
- Open category URLs like /categories/:categoryId.
- Open product pages by numeric ID or product slug.
- View product detail pages with gallery images, variants, pricing, stock, ratings, related products, and reviews.
- Add products or variants to cart.
- Add/remove products from wishlist.
- See polished empty states for no products, no wishlist items, no orders, and no notifications.
- Use light/dark mode.

Accounts And Security

Users can:

- Register and log in.
- Verify email through an email verification flow.
- Use password reset and forgot-password flows.
- Update profile details.
- Change password.
- Manage active sessions.
- Log out.

- Use optional two-factor authentication support.
- Access protected pages only when authenticated.

Admins are separated by role-based access through protected admin routes.

Cart And Checkout

The checkout flow supports:

- Cart page with quantity changes and item removal.
- Checkout steps: cart, shipping, payment, confirmation.
- Saved shipping addresses.
- New address creation during checkout.
- Country/state dropdown-style address fields.
- Postal-code validation by country.
- Map picker support with better feedback.
- Default address selection directly during checkout.
- Standard and express shipping.
- Coupon application.
- Draft checkout persistence in browser storage.
- Detection when cart contents change during checkout.
- “Return to cart” behavior that clearly explains it cancels the unpaid checkout attempt while keeping cart items.
- Edit-shipping flow after checkout has started, before payment capture.
- Stripe Payment Element support when Stripe keys are configured.
- Demo payment mode when Stripe is not configured.
- Better failed/declined payment messaging and retry handling.

Orders

Customers can:

- View their order history.
- Open order detail pages.
- See order items, totals, discount, tax, shipping, and final amount.
- Track order status.

- See timeline/progress.
- Cancel eligible orders before fulfillment.
- Receive order confirmation and shipment emails.

Supported order statuses include:

- AWAITING_PAYMENT
- PENDING
- CONFIRMED
- PACKED
- SHIPPED
- DELIVERED
- CANCELLED
- REFUNDED

Admin Dashboard

The admin dashboard includes:

- Total orders.
- Pending orders.
- Low-stock product count.
- Revenue today.
- Total revenue.
- Average order value.
- Revenue and order-count charts by day, week, or month.
- Best-selling product visual rankings.
- Low-stock/high-demand visual rankings.
- New customer trend chart.
- Low-stock visual meters.
- Quick +10 restock action.
- Recent order flow/status visualization.
- Editable low-stock threshold.

Long list-heavy dashboard sections have been replaced with more readable visual panels.

Admin Products

Admins can:

- **Create products.**
- **Edit products.**
- **Save products as draft.**
- **Publish products.**
- **Mark products featured.**
- **Assign categories.**
- **Set SKU.**
- **Set slug.**
- **Set meta title and meta description.**
- **Set visibility dates.**
- **Set base stock quantity.**
- **Manage variants with size, color, SKU, stock, price override, and active state.**
- **Upload product images.**
- **Add multiple gallery images.**
- **Reorder gallery images.**
- **Mark images as primary.**
- **Preview gallery images.**
- **Delete images.**
- **Search product table by name, SKU, or category.**
- **Filter by status: all, published, draft, low stock.**
- **Filter by category.**
- **Use responsive mobile product cards instead of cramped tables.**
- **Manually adjust inventory and view recent stock adjustment history.**

Admin Categories

Admins can:

- **Create categories.**
- **Edit categories.**

- Delete categories.
- Use categories for storefront filtering and product organization.

Admin Orders And Fulfillment

Admins can:

- View all orders.
- Search orders by ID, customer name, or customer email.
- Filter by status.
- Use responsive mobile order cards.
- Open detailed admin order pages.
- Update fulfillment status.
- Mark orders packed, shipped, or delivered.
- Add shipping carrier.
- Add tracking number.
- Generate tracking URL behavior.
- Add admin notes.
- Add timeline notes.
- View staff timeline events.
- Download invoice PDF.
- Process partial refunds.
- Track refunded and remaining refundable amounts.
- See refund history and refund reasons.

Admin Reports And Exports

Admins can export:

- Orders CSV.
- Customers CSV.
- Products/inventory CSV.
- Invoice/receipt PDF by order ID.

Reports can be filtered by:

- Date range.

- Order status.
- Customer status.
- Product status, including active, inactive, and low stock.

Coupons And Promotions

Admins can:

- Create coupons.
- Edit coupons.
- Delete coupons.
- Configure fixed or percent discounts.
- Configure minimum order amount.
- Configure active/inactive state.
- Configure expiration.
- Configure usage limits.

Customers can apply coupons during checkout, including free-shipping style discounts when configured.

Notifications

The app includes:

- Notification list.
- Recent notifications.
- Unread count.
- Mark one notification as read.
- Mark all as read.
- Clear notifications.
- Notification settings.
- Real-time notification updates through SSE.
- Click-through behavior to relevant order/product areas when applicable.

Store Settings And Branding

Admins can configure:

- Brand name.

- Logo URL.
- Contact email.
- Contact phone.
- Contact address.
- Homepage banner title.
- Homepage banner subtitle.
- Homepage banner image URL.
- Homepage CTA text.
- Homepage CTA URL.
- Tax rate.
- Standard shipping cost.
- Express shipping cost.
- Low-stock threshold.

The storefront uses these settings for branding, homepage content, checkout pricing, and email templates.

Trust And Commercial Pages

Public pages include:

- About page.
- Contact page.
- Privacy policy.
- Return policy.
- Shipping policy.
- Terms and conditions.
- Footer links to policies and support pages.

Email System

The backend has branded email templates for:

- Order confirmation.
- Shipment notification.
- Password reset.

- Email verification.

When mail is disabled, email HTML is logged for local development. When configured, it uses SMTP.

Images And Storage

The app supports:

- Admin image upload.
- Local file storage served from /uploads/**.
- Cloudinary storage option.
- Product main images from locally generated/uploaded assets.
- Seeded demo product images generated into the local upload folder.
- External image URL support where used.

Demo Data

Startup seed support can create a full demo store:

- Store branding/settings.
- Categories.
- Products.
- Variants.
- SKUs.
- Inventory levels.
- Multiple product images.
- Customers.
- Addresses.
- Default addresses.
- Wishlists.
- Reviews.
- Coupons.
- Orders across many statuses.
- Refunds.
- Tracking data.
- Notifications.

- Admin account.

Use this to show buyers the full platform without manually entering data.

Technical/Developer Features

Backend includes:

- Spring Boot 4.
- Spring Security with JWT.
- Role-based access control.
- JPA entities and repositories.
- PostgreSQL support.
- H2 test database.
- Stripe payments and webhook endpoint.
- OpenAPI/Swagger docs.
- Validation and global error handling.
- Audit logging.
- Request logging.
- Async email sending.
- Scheduled abandoned order cleanup.
- Local and Cloudinary file storage options.

Frontend includes:

- React 19.
- Vite.
- MUI.
- React Router.
- Stripe Elements.
- Leaflet map support.
- Toast notifications.
- Shared API client with better error messages.
- Responsive admin layouts.
- Loading skeletons and polished empty/error states.

Current Buyer Pitch

This is a modern commerce starter platform for small-to-mid-size online stores. It covers the important selling points a buyer expects: storefront, catalog search/filtering, checkout, Stripe payments, customer accounts, order tracking, admin operations, product variants, inventory, refunds, coupons, reporting/export, invoices, branded emails, store settings, and demo data for evaluation.

The strongest areas are admin operations, checkout UX, product management, reporting, and commercial polish. The main production work before a real launch would be deployment hardening, final payment/webhook QA with real Stripe keys, real SMTP configuration, legal copy review, and any client-specific branding/design pass.